

Solution Q.1 ES-341

October 2, 2026

Problem 01

Question: Find the fourth-order Taylor polynomial $P_4(x)$ for the function $f(x) = xe^{x^2}$ about $x = 0$ by truncating the Taylor series to the 5th term.

Solution:

The standard Taylor series expansion for e^u centered at $u = 0$ is given by:

$$e^u = \sum_{n=0}^{\infty} \frac{u^n}{n!} = 1 + u + \frac{u^2}{2!} + \frac{u^3}{3!} + \cdots$$

Substituting $u = x^2$ into the expansion yields:

$$e^{x^2} = 1 + x^2 + \frac{x^4}{2} + \frac{x^6}{6} + \cdots$$

Multiplying the entire series by x gives the series representation of $f(x) = xe^{x^2}$:

$$f(x) = x \left(1 + x^2 + \frac{x^4}{2} + \frac{x^6}{6} + \cdots \right) = x + x^3 + \frac{x^5}{2} + \frac{x^7}{6} + \cdots$$

Note that the Taylor series of $f(x)$ contains only odd powers of x :

- 1st term: x
- 2nd term: x^3
- 3rd term: $\frac{x^5}{2}$
- 4th term: $\frac{x^7}{6}$
- 5th term: $\frac{x^9}{24}$

Truncating the series to the 5th term gives:

$$T_5(x) = x + x^3 + \frac{x^5}{2} + \frac{x^7}{6} + \frac{x^9}{24}$$

However, the degree N Taylor polynomial $P_N(x)$ contains all terms of degree $\leq N$. For $N = 4$, since the coefficient of x^4 in the Maclaurin expansion of $f(x)$ is 0, the fourth-order Taylor polynomial $P_4(x)$ consists of all powers up to x^4 :

$$P_4(x) = x + x^3$$

Problem 02

Question: How many bisections of the interval $(0, 1)$ are required to find the root of $f(x) = x - 2^{-x}$, correct to 10^{-5} order?

Solution:

For the Bisection Method on an initial interval $[a, b]$, the error bound after n iterations is given by:

$$\text{Error}_n = \frac{b - a}{2^n}$$

We are given:

- Initial interval: $[a, b] = [0, 1] \implies b - a = 1 - 0 = 1$
- Desired accuracy: $\text{Error}_n \leq 10^{-5}$

Setting up the inequality:

$$\frac{1}{2^n} \leq 10^{-5}$$

Taking the reciprocal on both sides reverses the inequality:

$$2^n \geq 10^5 = 100,000$$

Taking the natural logarithm (or base-10 logarithm) on both sides:

$$\begin{aligned} n \ln(2) &\geq \ln(10^5) = 5 \ln(10) \\ n &\geq \frac{5 \ln(10)}{\ln(2)} \approx \frac{5 \times 2.302585}{0.693147} \approx 16.6096 \end{aligned}$$

Since the number of bisections n must be an integer, we round up to the next integer:

$$n = \lceil 16.6096 \rceil = 17$$

Conclusion: A minimum of **17 bisections** are required to guarantee a tolerance of 10^{-5} .